Thursday Warm Up: Analog-to-Digital Conversion (ADC)

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1 Analog-to-Digital Conversion

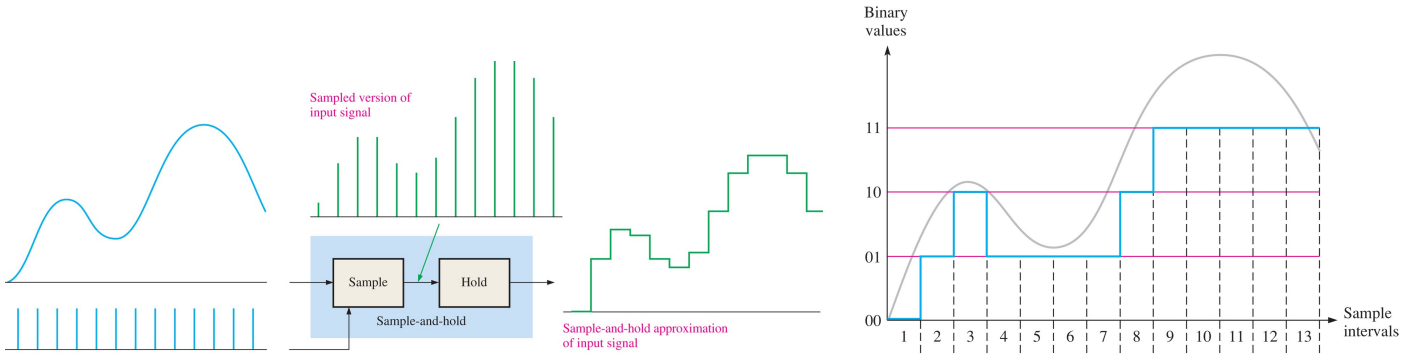


Figure 1: (Left) The sample-and-hold process. (Right) The waveform after quantization with four levels.

1. Imagine a circuit that mixes a train of impulses (Fig. 1 (left)) with an analog waveform, then holds the voltage constant between impulses. (a) If we had an *analog comparator* that was HIGH if each constant voltage was greater than a reference voltage, and LOW if it was not, then we could translate the constant voltages into binary numbers. If the code 11 in Fig. 1 (right) represents 3V, and 00 represents 0V, what is the voltage resolution? (b) What is true about the waveform for sampling intervals 9-13?

2. In Fig. 2 (left), $V_{\rm ref} = +8$ Volts, and the op-amp comparators act as the *analog comparators* mentioned in exercise 1. (a) Determine the binary code output of the 3-bit flash ADC in Fig. 2 (left), given the input signal in Fig. 1 (right), given the encoder enable pulses shown. (b) If the enable pulse frequency in Fig. 2 (right) were halved, determine the binary numbers represented by the resulting digital output. Is any information lost?

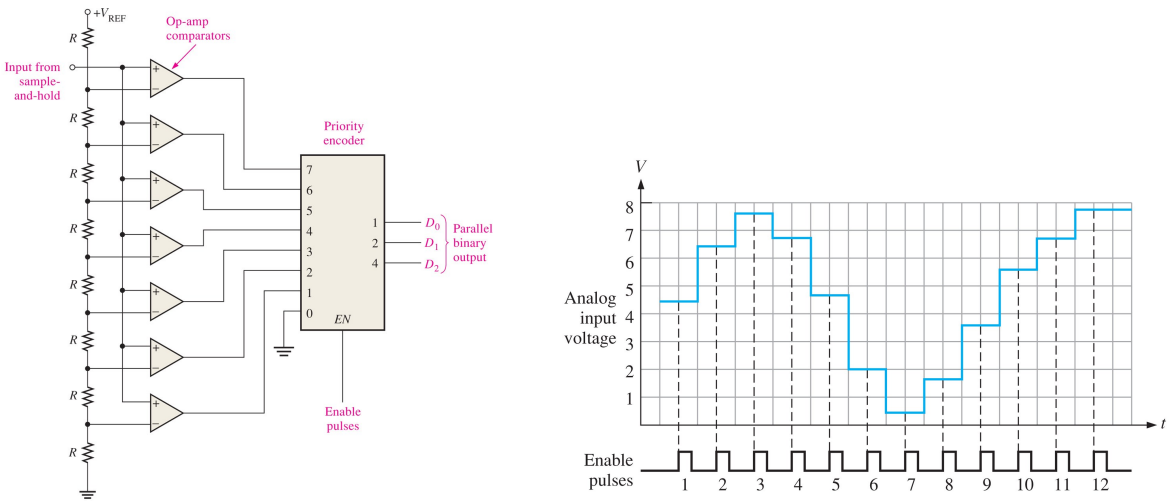


Figure 2: (Left) The 3-bit **Flash ADC**. (Right) The input to the 3-bit Flash ADC.